

Unconstrained Necessary Conditions

Learning objectives

After studying this chapter, you should be able to

- derive the one-dimensional and multivariate first-order necessary conditions for an interior local optimizer;
- distinguish a stationary point from a local or global optimizer;
- prove and apply the second-order necessary condition;
- explain, using Taylor expansion, why a positive semidefinite Hessian is inconclusive; and
- use stationary-point enumeration correctly for global conclusions, including its existence and finiteness requirements.

Previous lectures established the global framework of optimization: when minimizers exist, how local and global optimality differ, and why convex problems admit exact optimality certificates. We now remove convexity and ask a more delicate question:

Can derivatives at one point characterize an optimizer of a general smooth function?

The answer is asymmetric. Derivatives give conditions that every smooth interior optimizer must satisfy, but those conditions are generally not sufficient. They generate and reject candidates; additional structure is needed to certify optimality.

Throughout the chapter, let $\mathcal{D} \subseteq \mathbb{R}^n$ be open and consider

$$\min_{x \in \mathcal{D}} f(x).$$

Openness means that every sufficiently short displacement from a point of $\mathcal{D}$, in every direction, remains in the domain. This two-sided freedom is exactly what produces the unconstrained gradient condition.

6.1 First-order necessary condition: one-dimensional cases

We first isolate the elementary one-dimensional argument.

Theorem 6.1 (Fermat's necessary condition in one dimension). *Let $\mathcal{D} \subseteq \mathbb{R}$ be open, let $x^* \in \mathcal{D}$, and suppose $\phi : \mathcal{D} \rightarrow \mathbb{R}$ is differentiable at x^* . If x^* is a local minimizer or a local maximizer of ϕ , then*

$$\phi'(x^*) = 0.$$

Proof. Suppose first that x^* is a local minimizer. For every sufficiently small h ,

$$\phi(x^* + h) - \phi(x^*) \geq 0.$$

Consequently,

$$\frac{\phi(x^* + h) - \phi(x^*)}{h} \geq 0 \quad \text{when } h > 0,$$

whereas

$$\frac{\phi(x^* + h) - \phi(x^*)}{h} \leq 0 \quad \text{when } h < 0.$$

Differentiability makes both one-sided limits equal to $\phi'(x^*)$. The right-hand limit implies $\phi'(x^*) \geq 0$, while the left-hand limit implies $\phi'(x^*) \leq 0$. Thus $\phi'(x^*) = 0$. If x^* is a local maximizer, apply the minimum result to $-\phi$. $\square$

The proof does not require the derivative to change sign strictly. For example, a function may be constant on an interval. What matters is that the same derivative must be compatible with the nonnegative right difference quotients and the nonpositive left difference quotients.

6.2 First-order necessary condition: multi-dimensional cases

Every direction through a multivariate point produces a one-dimensional restriction. For $\mathbf{d} \in \mathbb{R}^n$, define

$$\phi_{\mathbf{d}}(t) := f(\mathbf{x}^* + t\mathbf{d}).$$

Theorem 6.2 (First-order necessary condition). *Let $\mathcal{D} \subseteq \mathbb{R}^n$ be open, let $\mathbf{x}^* \in \mathcal{D}$, and suppose $f : \mathcal{D} \rightarrow \mathbb{R}$ is differentiable at $\mathbf{x}^*$. If $\mathbf{x}^*$ is a local minimizer or a local maximizer of f , then*

$$\nabla f(\mathbf{x}^*) = \mathbf{0}.$$

Proof. Suppose $\mathbf{x}^*$ is a local minimizer. Fix any $\mathbf{d} \in \mathbb{R}^n$. Because $\mathcal{D}$ is open, $\phi_{\mathbf{d}}$ is defined for all sufficiently small positive and negative t , and $t = 0$ is a local minimizer of $\phi_{\mathbf{d}}$. [Theorem 6.1](#) therefore gives

$$0 = \phi'_{\mathbf{d}}(0) = \nabla f(\mathbf{x}^*)^{\top} \mathbf{d}.$$

This identity holds for every $\mathbf{d}$. In particular, choosing $\mathbf{d} = \nabla f(\mathbf{x}^*)$ gives $\|\nabla f(\mathbf{x}^*)\|^2 = 0$, so $\nabla f(\mathbf{x}^*) = \mathbf{0}$. Apply the result to $-f$ for a local maximizer. $\square$

Definition 6.1 (Stationary point and saddle point). A point $\mathbf{x}^* \in \mathcal{D}$ satisfying

$$\nabla f(\mathbf{x}^*) = \mathbf{0}$$

is a *stationary point* of f . A stationary point that is neither a local minimizer nor a local maximizer is called a *saddle point*.

Warning

These roles need not be mutually exclusive. If f is locally constant near $\mathbf{x}^*$, then $\mathbf{x}^*$ is both a local minimizer and a local maximizer. Under the definition above it is not a saddle point.

Proposition 6.3 (Stationary roles on an open domain). *Let $\mathcal{D} \subseteq \mathbb{R}^n$ be open, and let*

$f : \mathcal{D} \rightarrow \mathbb{R}$ be differentiable. Define

$$\begin{aligned}\mathcal{S} &:= \{\mathbf{x} \in \mathcal{D} : \nabla f(\mathbf{x}) = \mathbf{0}\}, \\ \mathcal{L}_{\min} &:= \{\mathbf{x} \in \mathcal{D} : \mathbf{x} \text{ is a local minimizer of } f\}, \\ \mathcal{L}_{\max} &:= \{\mathbf{x} \in \mathcal{D} : \mathbf{x} \text{ is a local maximizer of } f\}, \\ \mathcal{G}_{\min} &:= \arg \min_{\mathbf{x} \in \mathcal{D}} f(\mathbf{x}), \quad \mathcal{G}_{\max} := \arg \max_{\mathbf{x} \in \mathcal{D}} f(\mathbf{x}).\end{aligned}$$

Then

$$\mathcal{G}_{\min} \subseteq \mathcal{L}_{\min} \subseteq \mathcal{S}, \quad \mathcal{G}_{\max} \subseteq \mathcal{L}_{\max} \subseteq \mathcal{S}.$$

If $\mathcal{G}_{\min} \cap \mathcal{G}_{\max} \neq \emptyset$, then f is constant on $\mathcal{D}$, and all five sets equal $\mathcal{D}$. An illustration can be found in [Figure 6.1](#).

Proof. Every global optimizer is a local optimizer. Every local optimizer is stationary by [Theorem 6.2](#), which proves the two chains of inclusions. If $\mathbf{x}^* \in \mathcal{G}_{\min} \cap \mathcal{G}_{\max}$, then for every $\mathbf{x} \in \mathcal{D}$,

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \leq f(\mathbf{x}^*).$$

Hence $f(\mathbf{x}) = f(\mathbf{x}^*)$ throughout $\mathcal{D}$. Conversely, if f is constant, every point is simultaneously stationary, locally minimizing and maximizing, and globally minimizing and maximizing. $\square$

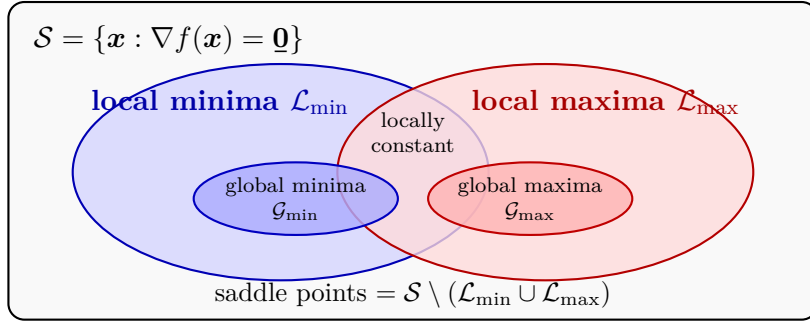


Figure 6.1: The stationary set and the possible local and global roles when $\mathcal{G}_{\min} \cap \mathcal{G}_{\max} = \emptyset$. The overlaps represent locally constant behavior. If $\mathcal{G}_{\min} \cap \mathcal{G}_{\max} \neq \emptyset$, the function is a constant on its domain.

Example 6.1 (Solving the gradient equation gives candidates). Let

$$f(x_1, x_2) := x_1^3 - 3x_1 + x_2^2.$$

Since

$$\nabla f(x_1, x_2) = \begin{bmatrix} 3x_1^2 - 3 \\ 2x_2 \end{bmatrix},$$

the stationary points are $(1, 0)^\top$ and $(-1, 0)^\top$. At this stage the calculation proves only that every interior local optimizer must belong to this list.

Example 6.2 (Stationarity allows every local role). At $x^* = 0$, each of x^2 , $-x^2$, and x^3 has zero derivative. Nevertheless, 0 is respectively a global minimum, a global maximum, and a saddle point.

Stationary points can also be local but not global. For

$$\phi(x) := x^2 - x^4 = x^2(1 - x^2),$$

we have $\phi(x) \geq \phi(0)$ whenever $|x| < 1$, but $\phi(x) \rightarrow -\infty$ as $|x| \rightarrow \infty$. Thus 0 is a strict local minimum but not a global minimum. Similarly, $-x^2 + x^4$ has a strict local maximum at 0 but is unbounded above.

Proposition 6.4 (Stationary roles under convexity). *Let $\mathcal{D} \subseteq \mathbb{R}^n$ be nonempty, open, and convex, and let $f : \mathcal{D} \rightarrow \mathbb{R}$ be differentiable and convex. Then*

$$\mathcal{S} = \mathcal{L}_{\min} = \mathcal{G}_{\min}, \quad \mathcal{L}_{\max} = \text{int}_{\mathcal{D}}(\mathcal{G}_{\min}) \subseteq \mathcal{G}_{\min},$$

where $\text{int}_{\mathcal{D}}$ denotes interior relative to $\mathcal{D}$. Consequently, f has no saddle points. Moreover,

$$\mathcal{G}_{\max} \neq \emptyset \iff f \text{ is constant on } \mathcal{D}.$$

In the constant case, all five sets in Proposition 6.3 equal $\mathcal{D}$.

Proof. The first inclusion chain in Proposition 6.3 gives $\mathcal{G}_{\min} \subseteq \mathcal{L}_{\min} \subseteq \mathcal{S}$. Conversely, if $\mathbf{x}^* \in \mathcal{S}$, the convex tangent-plane inequality gives, for every $\mathbf{x} \in \mathcal{D}$,

$$f(\mathbf{x}) \geq f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*) = f(\mathbf{x}^*).$$

Thus $\mathbf{x}^* \in \mathcal{G}_{\min}$, proving the equality of the first three sets.

Now let $\mathbf{x}^* \in \mathcal{L}_{\max}$. The first-order necessary condition gives $\mathbf{x}^* \in \mathcal{S} = \mathcal{G}_{\min}$. Local maximality provides a neighborhood U of $\mathbf{x}^*$, relative to $\mathcal{D}$, on which $f(\mathbf{x}) \leq f(\mathbf{x}^*)$. Global minimality gives the reverse inequality, so f is constant on U . Hence $\mathbf{x}^* \in \text{int}_{\mathcal{D}}(\mathcal{G}_{\min})$. Conversely, every point in this relative interior has a neighborhood on which f is constant and is therefore a local maximizer. This proves the asserted equality for $\mathcal{L}_{\max}$. Since every stationary point is a local minimizer, the saddle set is empty.

If $\mathbf{x}^* \in \mathcal{G}_{\max}$, then $\mathbf{x}^* \in \mathcal{L}_{\max} \subseteq \mathcal{G}_{\min}$. Hence $\mathcal{G}_{\min} \cap \mathcal{G}_{\max} \neq \emptyset$, and Proposition 6.3 implies that f is constant. Conversely, a constant function attains its global maximum at every point of the nonempty domain. $\square$

Example 6.3 (A flat minimum set and the constant case). Consider

$$p(x) := (\max\{|x| - 1, 0\})^2, \quad x \in \mathbb{R}.$$

It can be written as

$$p(x) = \begin{cases} (x+1)^2, & x < -1, \\ 0, & -1 \leq x \leq 1, \\ (x-1)^2, & x > 1. \end{cases}$$

Its derivative is

$$p'(x) = \begin{cases} 2(x+1), & x < -1, \\ 0, & -1 \leq x \leq 1, \\ 2(x-1), & x > 1. \end{cases}$$

The derivative is continuous and nondecreasing, so p is differentiable and convex. Directly from the formulas,

$$\mathcal{S} = \mathcal{L}_{\min} = \mathcal{G}_{\min} = [-1, 1], \quad \mathcal{L}_{\max} = (-1, 1), \quad \mathcal{G}_{\max} = \emptyset.$$

Thus an interior point of the flat minimum set is simultaneously a global minimizer and a non-strict local maximizer, even though p is not constant. The endpoints are not local maximizers because every neighborhood of either endpoint contains points where $p > 0$. By contrast, if $g(x) \equiv c$, then every point is stationary and is simultaneously a local and global minimizer and maximizer.

6.3 Second-order necessary conditions

First-order stationarity ignores curvature. A second derivative rejects additional candidates, but the resulting condition is still one-way.

Theorem 6.5 (First- and second-order necessary conditions). *Let $\mathcal{D} \subseteq \mathbb{R}^n$ be open, let $\mathbf{x}^* \in \mathcal{D}$, and suppose $f \in C^2$ near $\mathbf{x}^*$.*

(i) *If $\mathbf{x}^*$ is a local minimizer, then*

$$\nabla f(\mathbf{x}^*) = \mathbf{0}, \quad \mathbf{H}_f(\mathbf{x}^*) \succeq 0.$$

(ii) *If $\mathbf{x}^*$ is a local maximizer, then*

$$\nabla f(\mathbf{x}^*) = \mathbf{0}, \quad \mathbf{H}_f(\mathbf{x}^*) \preceq 0.$$

Proof. The gradient statements follow from [Theorem 6.2](#). Suppose $\mathbf{x}^*$ is a local minimizer and fix $\mathbf{d} \in \mathbb{R}^n$. For $\phi_{\mathbf{d}}(t) = f(\mathbf{x}^* + t\mathbf{d})$, the point 0 is a local minimizer and $\phi'_{\mathbf{d}}(0) = 0$. Taylor expansion at 0 gives

$$\phi_{\mathbf{d}}(t) - \phi_{\mathbf{d}}(0) = \frac{1}{2}t^2\phi''_{\mathbf{d}}(0) + o(t^2).$$

If $\phi''_{\mathbf{d}}(0) < 0$, the right-hand side is negative for all sufficiently small nonzero t , contradicting local minimality. Hence

$$\mathbf{d}^\top \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} = \phi''_{\mathbf{d}}(0) \geq 0 \quad \text{for every } \mathbf{d} \in \mathbb{R}^n.$$

This is exactly $\mathbf{H}_f(\mathbf{x}^*) \succeq 0$. Apply the minimum result to $-f$ to obtain the maximum statement. $\square$

Example 6.4 (Identical first- and second-order data). For

$$\begin{aligned} \phi_1(x) &= x^4, & \phi_2(x) &= -x^4, \\ \phi_3(x) &= x^4 - x^5, & \phi_4(x) &= -x^4 + x^5, \\ \phi_5(x) &= x^3. \end{aligned}$$

all five functions satisfy

$$\phi'_i(0) = 0, \quad \phi''_i(0) = 0.$$

The point 0 is a strict global minimum of ϕ_1 , a strict global maximum of ϕ_2 , and a saddle point of ϕ_5 . Moreover,

$$\phi_3(x) = x^4(1 - x) > 0 = \phi_3(0) \quad (0 < |x| < 1),$$

so 0 is a strict local minimum of ϕ_3 . It is not global because $\phi_3(x) \rightarrow -\infty$ as $x \rightarrow +\infty$. Applying the same argument to $\phi_4 = -\phi_3$ shows that 0 is a strict local maximum of ϕ_4 ,

but not a global maximum. Thus identical first- and second-order data can lead to five different roles, and the combined necessary conditions remain insufficient.

Example 6.5 (A positive-semidefinite Hessian can hide a saddle). Consider

$$f(x_1, x_2) := x_1^2 - x_2^4.$$

At the origin,

$$\nabla f(\mathbf{0}) = \mathbf{0}, \quad \mathbf{H}_f(\mathbf{0}) = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \succeq 0.$$

Nevertheless,

$$f(t, 0) = t^2 > 0, \quad f(0, t) = -t^4 < 0 \quad (t \neq 0).$$

Every neighborhood of the origin contains larger and smaller values, so the origin is a saddle point.

Warning

The condition $\mathbf{H}_f(\mathbf{x}^*) \succeq 0$ concerns one point and does not imply that f is convex. The Hessian characterization of convexity requires $\mathbf{H}_f(\mathbf{x}) \succeq 0$ at every point of an open convex domain.

6.4 Missing higher-order information

At a stationary point, second-order Taylor expansion has the form

$$f(\mathbf{x}^* + \mathbf{d}) - f(\mathbf{x}^*) = \frac{1}{2} \mathbf{d}^\top \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + o(\|\mathbf{d}\|^2).$$

If $\mathbf{H}_f(\mathbf{x}^*) \succeq 0$ is singular, there is a nonzero direction $\mathbf{d}$ with

$$\mathbf{d}^\top \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} = 0.$$

Along this direction the quadratic term vanishes, and the sign of the function change is determined by information discarded in the second-order truncation. [Example 6.4](#) shows that the omitted terms can produce five different local and global roles despite identical first- and second-order data.

The issue is the absence of a *uniform positive curvature margin*. If $\mathbf{H}_f(\mathbf{x}^*) \succ 0$, then some $m > 0$ satisfies

$$\mathbf{d}^\top \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} \geq m \|\mathbf{d}\|^2 \quad \text{for every } \mathbf{d}.$$

The quadratic term can then dominate the Taylor remainder. [Chapter 7](#) turns this observation into a sufficient condition for a strict local minimum.

Convexity resolves the problem in a different way: its tangent-plane inequality is an exact comparison over the whole domain, rather than a local Taylor truncation. As [Proposition 6.4](#) shows, that inequality turns $\nabla f(\mathbf{x}^*) = \mathbf{0}$ into a global certificate by comparing $f(\mathbf{x}^*)$ with every feasible point.

6.5 Stationary-point enumeration and its limits

Necessary conditions can support a global search when existence and exhaustion are established separately.

Corollary 6.6 (Existence and a finite stationary set). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuously differentiable and coercive, and suppose its stationary set*

$$\mathcal{S} := \{\mathbf{x} \in \mathbb{R}^n : \nabla f(\mathbf{x}) = \mathbf{0}\}$$

is finite. Then $\mathcal{S}$ is nonempty, and the global minimizers of f are exactly

$$\arg \min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x}).$$

Proof. Continuity and coercivity imply that f has a global minimizer $\mathbf{x}^*$. Because the domain is the open set $\mathbb{R}^n$, [Theorem 6.2](#) gives $\nabla f(\mathbf{x}^*) = \mathbf{0}$. Thus $\mathbf{x}^* \in \mathcal{S}$, so $\mathcal{S}$ is nonempty.

Since $\mathcal{S}$ is finite, some $\hat{\mathbf{x}} \in \mathcal{S}$ minimizes f over $\mathcal{S}$. Global optimality of $\mathbf{x}^*$ and membership $\mathbf{x}^* \in \mathcal{S}$ give

$$f(\mathbf{x}^*) \leq f(\hat{\mathbf{x}}) \leq f(\mathbf{x}^*).$$

Hence $\hat{\mathbf{x}}$ is globally minimizing. Conversely, every global minimizer is stationary by the first-order necessary condition and must attain the smallest value on $\mathcal{S}$. $\square$

Important note: A valid finite comparison

The logical order is

$$\begin{aligned} &\text{existence} + \text{candidate exhaustion} + \text{value comparison} \\ &\implies \text{global minimizers.} \end{aligned}$$

None of these three ingredients can be silently omitted.

Example 6.6 (A complete finite search). Let

$$f(x_1, x_2) := (x_1^2 - 1)^2 + (x_2 - 2)^2.$$

The function is continuous and coercive, so a global minimizer exists. Its gradient is

$$\nabla f(x_1, x_2) = \begin{bmatrix} 4x_1(x_1^2 - 1) \\ 2(x_2 - 2) \end{bmatrix}.$$

Therefore

$$\mathcal{S} = \{(-1, 2)^\top, (0, 2)^\top, (1, 2)^\top\}.$$

The objective values are

$$f(-1, 2) = 0, \quad f(0, 2) = 1, \quad f(1, 2) = 0.$$

[Corollary 6.6](#) shows that the global minimizers are $(-1, 2)^\top$ and $(1, 2)^\top$. At the remaining candidate,

$$f(t, 2) = (t^2 - 1)^2 < 1 = f(0, 2) \quad (0 < |t| < \sqrt{2}),$$

whereas

$$f(0, 2 + t) = 1 + t^2 > 1 = f(0, 2) \quad (t \neq 0).$$

Hence $(0, 2)^T$ is a saddle point directly from the definition.

Example 6.7 (A finite list without existence). For

$$\phi(x) := x^3 - 3x,$$

the stationary points are $x = \pm 1$, and $\phi(1) = -2 < 2 = \phi(-1)$. Nevertheless,

$$\phi(x) \longrightarrow -\infty \quad \text{as } x \longrightarrow -\infty.$$

Thus choosing $x = 1$ because it has the smaller stationary value would be incorrect: no global minimizer exists.

This failure of global existence does not prevent local optimality. For $0 < |h| < 1$,

$$\phi(1 + h) - \phi(1) = h^2(3 + h) > 0,$$

so 1 is a strict local minimizer. Similarly,

$$\phi(-1 + h) - \phi(-1) = h^2(h - 3) < 0,$$

so -1 is a strict local maximizer. In fact, $\phi(x) \rightarrow +\infty$ as $x \rightarrow +\infty$, so no global maximizer exists either.

Warning

Well-posedness and existence are needed when stationary-point comparison is used to draw a *global* conclusion. They are not prerequisites for local minimizers, local maximizers, or saddle points, whose definitions depend only on behavior in a neighborhood.

Example 6.8 (Infinitely many isolated strict local minima). Let

$$f(x) := \sin^2 x + 1 - \cos x, \quad x \in \mathbb{R}.$$

Then

$$f'(x) = \sin x(2 \cos x + 1),$$

so every $k\pi$, $k \in \mathbb{Z}$, is stationary. Since

$$f(x) = (1 - \cos x)(2 + \cos x) \geq 0,$$

the points $2k\pi$ are global minimizers. They are also strict local minimizers: for every nonzero h with $|h| < \pi$,

$$f(2k\pi + h) - f(2k\pi) = (1 - \cos h)(2 + \cos h) > 0.$$

At the odd multiples of π , for $0 < |h| < \pi/2$,

$$f((2k + 1)\pi + h) - f((2k + 1)\pi) = \cos h(1 - \cos h) > 0.$$

Thus every $(2k + 1)\pi$ is a strict local minimizer with value 2, but is not global. These points are isolated from one another, so even the set of isolated strict local minimizers can be infinite.

Warning

On an open domain, stationarity gives an exhaustive candidate set for optimizers that exist. It need not give a finite or analytically manageable candidate set, and comparing stationary values does not by itself prove existence.

Chapter summary

- An interior local optimizer of a differentiable function must be stationary.
- At an interior local minimum, the Hessian must be positive semidefinite; at a local maximum it must be negative semidefinite.
- A semidefinite Hessian is inconclusive because zero-curvature directions leave the sign to higher-order terms.
- For a global conclusion, stationary-point comparison is valid only after existence, interiority, and exhaustion of the candidate set have been secured.

6.6 Exercises

Exercise 6.1 (Why the domain matters). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = (x - 3)^2$. Analyze its restriction to each of the domains

$$\mathbb{R}, \quad (0, 1), \quad [0, 1].$$

For each domain, determine whether a global minimizer exists and whether the unconstrained first-order condition applies at a minimizer.

Exercise 6.2 (Stationary points in one dimension). Define $q : \mathbb{R} \rightarrow \mathbb{R}$ by

$$q(x) := \begin{cases} (x + 2)^4, & x < -2, \\ 0, & -2 \leq x \leq 1, \\ (x - 1)^4, & x > 1. \end{cases}$$

For each of the following six functions from $\mathbb{R}$ to $\mathbb{R}$, verify that 0 is stationary and determine whether it is a local minimizer, local maximizer, global minimizer, global maximizer, or saddle point:

$$q(x), \quad \frac{x^2}{1 + x^2}, \quad e^{-x^2}, \quad x^3 e^{-x^2}, \quad x^2 + x - e^x, \quad -x^2 + x + e^{-x}.$$

State every applicable category; the categories need not be mutually exclusive. Justify local classifications by direct comparisons or a sign chart, without using a second-order sufficient condition.

Exercise 6.3 (A zero Hessian can hide a saddle). Let $r : \mathbb{R}^2 \rightarrow \mathbb{R}$ be

$$r(x_1, x_2) = x_1^4 + x_2^4 - 3x_1^2 x_2^2.$$

Show that the origin passes both the local-minimum and local-maximum necessary-

condition tests. Then compare the values on the coordinate axes and on the line $x_1 = x_2$ to classify the origin.

Exercise 6.4 (Using the convex inequality). Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x_1, x_2) = e^{x_1} + e^{x_2} - x_1 - x_2.$$

Show that f is strictly convex, find its stationary point, and use the convex tangent-plane inequality to prove that this point is the unique global minimizer. Do not rely only on a Hessian value at the candidate.

Exercise 6.5 (A finite list is not an existence proof). Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be

$$\phi(x) = \frac{x^3}{1 + x^2}.$$

Find all stationary points and compare their values. Then prove that ϕ has no global minimizer and identify precisely which hypothesis of [Corollary 6.6](#) fails.

Exercise 6.6 (Infinitely many strict local minima). Let $\psi : \mathbb{R} \rightarrow \mathbb{R}$ be

$$\psi(x) := \sin x + \frac{x}{2}.$$

Find every stationary point. Using a sign chart for ψ' , show that ψ has infinitely many strict local minimizers and infinitely many strict local maximizers. Determine whether a global minimizer or global maximizer exists, and explain why a finite stationary-point comparison is unavailable.

Exercise 6.7 (A finite search with coupled variables). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be

$$f(x_1, x_2) = x_1^4 + x_2^4 - 2x_1x_2.$$

- (i) prove that a global minimizer exists;
- (ii) find every stationary point;
- (iii) classify the origin by comparing the restrictions to $x_2 = x_1$ and $x_2 = -x_1$;
- (iv) compare the stationary values and determine every global minimizer; and
- (v) verify the minimum value directly from

$$x_1^4 + x_2^4 \geq 2x_1^2x_2^2.$$

Exercise 6.8 (Classification directly from local comparisons). For $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$g(x_1, x_2) = x_1^2x_2 + x_2^3 - 3x_2,$$

find every stationary point. Classify the points $(\pm\sqrt{3}, 0)$ by restricting g to the vertical lines $x_1 = \pm\sqrt{3}$. At $(0, \pm 1)$, do not invoke Lecture 7: translate x_2 by ± 1 and use the resulting exact polynomial expansion to prove the local classification directly. Finally, decide whether g has a global minimizer or maximizer.

Exercise 6.9 (Integrated three-variable problem). For $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by

$$\begin{aligned} f(x_1, x_2, x_3) := & \log(x_1^2 + 1) + x_1^2 - 2x_1x_2 + 4x_2^2 \\ & + x_3^4 - 8x_3^3 + 16x_3^2, \end{aligned}$$

- (i) determine whether f is coercive;
- (ii) find every stationary point;
- (iii) apply every valid necessary-condition rejection test; and
- (iv) determine every global minimizer by a complete comparison argument.